# LangChain 2-Week Portfolio Roadmap

Day 2 Learning Summary & Practice Record (Roman Urdu)

Week 1 • Day 2

## 1. Aj Ka Topic: Dynamic Prompt Templates

Day 1 par humne seekha tha ke LLM ko simple static string prompt kaise bheja jata hai. Day 2 par humne seekha ke real-world AI applications mein static prompts nahi balkey **Prompt Templates** istemaal kiye jatay hain. Is se hum runtime par user se inputs (`topic`, `language`) le kar dynamic tareeqay se prompt tayar kar sakte hain.

## 2. Concepts & Key Definitions

| Concept | Wazahat (Explanation) |
|---|---|
| **PromptTemplate** | LangChain ka core tool jo raw text string ko dynamic placeholders `{variable}` ke sath wrap karta hai taake ise baar baar reuse kiya ja sake. |
| **input_variables** | Ek list/array hoti hai jo template ko batati hai ke kaun se parameters (jaise `topic`, `language`) runtime par fill hone hain. |
| **langchain_core** | LangChain ka modern modular architecture. Humne `from langchain_core.prompts import PromptTemplate` ka sahi import tareeqa apnaya. |
| **User Interaction** | Python ke `input()` function ka istemaal kar ke application ko fully dynamic aur interactive banaya gaya. |

## 3. Practical Execution Code (`app.py`)

```
import os from dotenv import load_dotenv from langchain_openai import ChatOpenAI from langchain_core.prompts import PromptTemplate load_dotenv() api_key = os.environ.get("OPEN_API_SECRET_KEY") # Initialize Chat Model llm = ChatOpenAI(model="gpt-4o-mini", temperature=0.7, api_key=api_key) # Prompt Template Setup prompt_template = PromptTemplate( input_variables=["topic", "language"], template="Mujhe {topic} ke baray mein 2 lines mein batao, aur iska jawab {language} mein hona chahiye." ) # User Interactive Input topic = input("Enter topic: ") language = input("Enter language: ") # Dynamic Formatting final_prompt = prompt_template.format(topic=topic, language=language) # Model Invocation response = llm.invoke(final_prompt) print(response.content)
```

### 💡 Day 2 Key Learning & Outcome

Hardcoded prompts flexibile nahi hotay. `PromptTemplate` ka asal faida reusability aur automated workflow hai, jisse dynamic user requirements ke mutabiq prompts generate kiye ja sakte hain.